A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Thermal & oil pollution - Set 1

Mark scheme

AQA spec 3.4.3.2.6-3.4.3.2.7; Pollution; textbook pp. 254-262

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	1 mark for the point below. **1** As temperature increases the maximum dissolved oxygen concentration decreases (non-linear: falls faster at low temperatures) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
01.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** About 39 °C (accept 38-40) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (°C) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1

Do not write outside the box

Question	Marking guidance	Marks
01.3	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 9.1 - 7.9 (value at 27 °C, accept 7.8-8.0) **1** = 1.2 mg dm^{-3} (accept 1.1-1.3) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (mg dm^{-3}) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Figure 1 shows the maximum / saturation values – processes that remove oxygen (aerobic decomposition of organic matter / respiration) can give values below the line **1** Higher temperature increases the metabolic rate of organisms and the rate of bacterial decomposition so oxygen is used faster – values above the line are impossible **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.5	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Increased metabolic rate so more energy used / less surplus for growth **1** Eggs develop faster / hatch earlier (out of step with food supply) **1** Non-indigenous warm-water species may thrive and out-compete indigenous species **1** Toxicity of other pollutants is greater as toxins are metabolised faster **1** Reduced resistance to disease **1** Proteins / enzymes denatured if temperature is very high (above about 40 °C) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
01.6	1 mark for the point below. **1** Hot water is sprayed from a shower floor in a tall hollow tower – air absorbs the heat and rises, drawing in more air (updraught), so the water returned is cooler (and saturated with oxygen) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 256-257; AQA spec 3.4.3.2.6.	

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Question	Marking guidance	Marks
02.1	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Discharge of tank-washing water from tankers **1** Oil rig / drilling accidents / leakage during drilling **1** Pipeline leaks **1** Leakage from storage tanks / refinery spills **1** Waste lubricating oil from vehicles / industry washed into drains **1** Other ship accidents (fuel oil) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
02.2	**1 mark each** 1 mark for each of the 3 points below. **1** Oil sticks feathers together so they lose their insulating properties and birds die of hypothermia **1** Feathers lose buoyancy / waterproofing so heavily oiled birds drown **1** Birds swallow toxic oil when preening – time spent cleaning feathers reduces time for feeding themselves / their young **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
02.3	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** The surface film reduces the diffusion of oxygen from the atmosphere into the water, so aerobic organisms suffer from low dissolved oxygen **1** Toxic components (benzene, xylene) dissolve and poison plankton near the surface **1** Oil blocks light so algae / phytoplankton cannot photosynthesise **1** Oil masks odours / narcotises so animals cannot find mates or food by smell **1** Shore organisms (molluscs, crustaceans) are covered and asphyxiated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

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Question	Marking guidance	Marks
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** Dispersal / dilution by currents and waves is prevented so oil stays concentrated – in open water most spills disperse in days or weeks **1** Inshore areas have more sensitive ecosystems (shores, reefs, mangroves, breeding sites) and more human activities (fishing, tourism) to be affected **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 258-259; AQA spec 3.4.3.2.7.	
03.1	**1 mark each** 1 mark for each of the 5 points below. **1 Double hull:** a gap of up to 2 m between the hulls so damage to the outer hull in a collision / grounding does not release oil **1 Inert gas system:** cooled engine exhaust gas (no oxygen) fills the space above the oil so an explosive oil vapour / air mixture cannot form **1 Separate ballast tanks:** ballast water is not carried in oil tanks so no residual oil is discharged when ballast is pumped out **1 Twin engines / rudders:** if one fails the ship can still be steered / propelled and is not carried onto rocks by currents or wind **1 Crude oil washing:** sludges are mixed and removed with the cargo so tanks do not need washing with seawater that was then discharged – oily water is unloaded at the terminal **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	5
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Offshore routes keep tankers further from the coast so if there is engine failure there is more time to deal with it before the ship is carried ashore **1** GPS gives accurate navigation – AIS lets ship positions be monitored and coordinated to avoid collisions and groundings **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accurate** → close to the true value	2

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Question	Marking guidance	Marks
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** The tank stands on an impermeable base surrounded by a wall **1** The volume enclosed by the wall is greater than the tank volume so all the oil is contained if the tank splits / cannot overflow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **impermeable** → does not let water through	2
03.4	1 mark for the point below. **1** Separates floating oil from drainage water so the oil is retained (for later treatment) and only water flows to the drains / river **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 259-260; AQA spec 3.4.3.2.7.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** Booms are floating tubes with a skirt that restrict the movement of floating oil, keeping it near the source / away from sensitive sites (mangroves, reef, marine park) **1** They work best in sheltered / calm water such as the lagoon **1 Limitation:** they do not work in open water where currents or waves are strong – oil can flow under the skirt in a strong current **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
04.2	**1 mark each** 1 mark for each of the 3 points below. **1 Skimmers:** rotating (metal) discs / belts pick up floating oil which is scraped off and stored **1** Absorbent materials (high surface area, eg textile mops, straw, human hair) soak up oil and are then removed **1** Collected oil / absorbents are incinerated or sent to landfill (or oil recycled) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3

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Question	Marking guidance	Marks
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Dispersants break the oil into an emulsion of droplets, making it more mobile so it spreads through the water column (to the reef / seabed organisms) rather than staying on the surface **1** They do not reduce the amount of oil – the dispersant / oil mixture may be toxic to corals and plankton in shallow, enclosed water **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** Bacteria that break down hydrocarbons digest the residual oil (most rapid where temperature, oxygen and nutrients are high) **1** Accelerated by adding nutrients (nitrates / phosphates) and aerating (ploughing) / irrigating the sediment **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 261-262; AQA spec 3.4.3.2.7.	
05.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (2900 - 1400) / 1400 x 100 **1** = 107% (accept 107.1) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Large decrease from 24 per year (1975-79) to 2 per year (2010-14) **1** Steepest fall between the late 1970s and 1980s – slower decline / levelling off after 1995 **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

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Question	Marking guidance	Marks
05.3	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Double-hulled tankers required for all large tankers **1** Improved navigation (GPS, AIS) and offshore shipping routes **1** Inert gas systems reducing explosions **1** Twin engines / rudders reducing groundings after mechanical failure **1** Stricter regulation, inspection and polluter-pays liability (eg after Exxon Valdez) **1** Separate ballast tanks / crude oil washing **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
05.4	**1 mark each** 1 mark for each of the 3 points below. **1** Most oil pollution comes from dispersed / diffuse sources not tanker accidents: waste lubricating oil, industrial oil in drains, road runoff **1 Routine operations:** small spills, pipeline leaks, storage tank leakage, oil rig accidents (eg Deepwater Horizon) **1** Growth in oil use and transport means more small releases even if large accidents are rarer **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **diffuse** → pollution from a wide area with no single outlet	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 258-262; AQA spec 3.4.3.2.7.	
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Steam washing (hot water / steam jets kill organisms in the sediment)	1
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Chemicals cause the oil molecules to join together / solidify **1** Producing more solid material that can be collected / lifted more easily than liquid oil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

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Question	Marking guidance	Marks
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Air contains about 210 cm^3 of oxygen per litre whereas water at 20 °C holds a maximum of about 6 cm^3 per litre (less than 3% of the amount in air) **1** So a small amount of oxygen-consuming activity (decomposition, respiration) or warming can remove most of the oxygen – many aquatic organisms are near the lower end of their range of tolerance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **deoxygenation** → loss of dissolved oxygen from the water • **range of tolerance** → the range of conditions in which a species can survive	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 254-262; AQA spec 3.4.3.2.6-3.4.3.2.7.	

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